

Community water deliveries outcomes: Description and methodology

Key outcome name: Community Surface Water Reliability

Additional data-in-depth outcomes: Monthly and annual community water deliveries, long-term running average of community water deliveries, M&I supply shortage and welfare loss from supply shortage.

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Background

The State Water Project serves 27 million residents, either by directly supplying their drinking water, or by recharging the groundwater basins that do so. The Central Valley project serves more than 6 million. Beyond providing drinking water, these deliveries also play a critical role in ensuring communities have water necessary to provide essential services, like running schools and hospitals, as well as maintaining parks. As the climate changes, California's water supply availability will change in quantity and timing, potentially putting deliveries to community water systems and wholesale providers at risk. We use the CalSim3 model to evaluate the impact of different water management strategies and hydrologic conditions on community water deliveries, often referred to as Municipal and Industrial, or M&I, demands.

M&I users include both community water systems (defined by the Safe Drinking Water Act as water systems serving 15 or more connections or 25 or more residents year round including water systems run by cities, special districts and Investor Owned Utilities) and non-public water system wholesale drinking water providers (special districts that sell water to community water systems but are not regulated as public water systems because they do not sell to any retail customers, notably these entities are regulated as urban water suppliers).

For this key outcome, the outcome levels indicate the reliability of these imported water supplies, based on how frequently a percentage of recent demand is met. The results provide information about the conditions under which reductions to these drinking water supplies might occur and the severity of any such reductions. The proceeding text documents the methods behind this key outcome which is evaluated at 74 locations of interest (LOIs), representing CalSim3 demand units and export deliveries that leave the model's geographic area. The document then describes two additional scenario outcomes related to community water supplies: M&I supply shortage and welfare loss from supply shortage.

Locations of Interest and associated water systems

In CalSim3, water demands are represented by demand units, which are assigned as agricultural or M&I. Additionally, some additional M&I water demands located outside of the model's geographic domain are represented as M&I diversion nodes. In collaboration with the COEQWAL modeling team, we identified 56 demand units and 18 diversion nodes with surface water deliveries modeled in CalSim3 where we could also confirm direct potable use of these surface water deliveries by one or more community water systems (see methodology step 1 below). Thus we exclude M&I demand units designated in CalSim3 as having exclusively groundwater pumping (e.g. the unit '62_NU') and M&I demand units and diversion nodes where water system reporting indicates that deliveries are utilized exclusively for conjunctive purposes including groundwater recharge and Aquifer Storage and Recovery (ASR) (e.g. the unit 'ESB415'). This resulted in 74 units for analysis as displayed in the ["master list of community surface water delivery locations of interest"](#).

To geographically display results for these locations of interest (LOIs) we use points representing the centroid for the service area of the largest system served (by population) by a demand unit or diversion node. We use this approach because some M&I diversion nodes do not have a spatial representation in the CalSim3 model because they fall outside of the model's geographic domain. A map of these LOIs is provided in Figure 1.

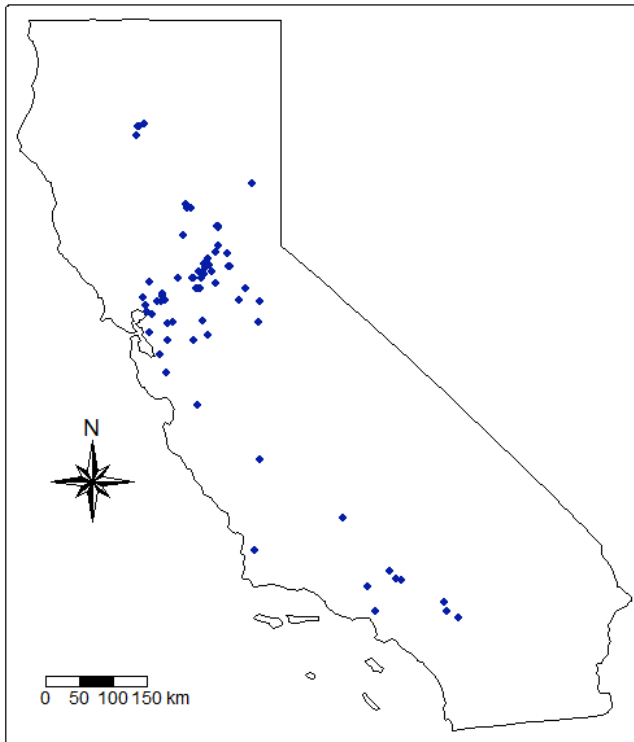


Figure 1. Distribution of 74 locations of interest analyzed for the community surface water reliability key outcome.

All results are presented at the LOI level and therefore do not directly correspond to results for any specific community water system or wholesale provider. However, to aid users in identifying results of particular relevance for specific communities, we also identify all of the California-regulated community water systems that utilize surface water deliveries associated with these LOIs (see methodology step 1 below). These community water systems are listed by LOI in the [“master list of systems served by community surface water delivery locations of interest”](#) datasheet (note that in this datasheet there can be multiple rows per water system if that water system is associated with more than one LOI). There are 475 unique community water systems in this list. These systems are mapped in Figure 2 using the centroid of their service boundary as identified in the State Water Resources Control Board’s System Boundary Area Layer (SABL) dataset.

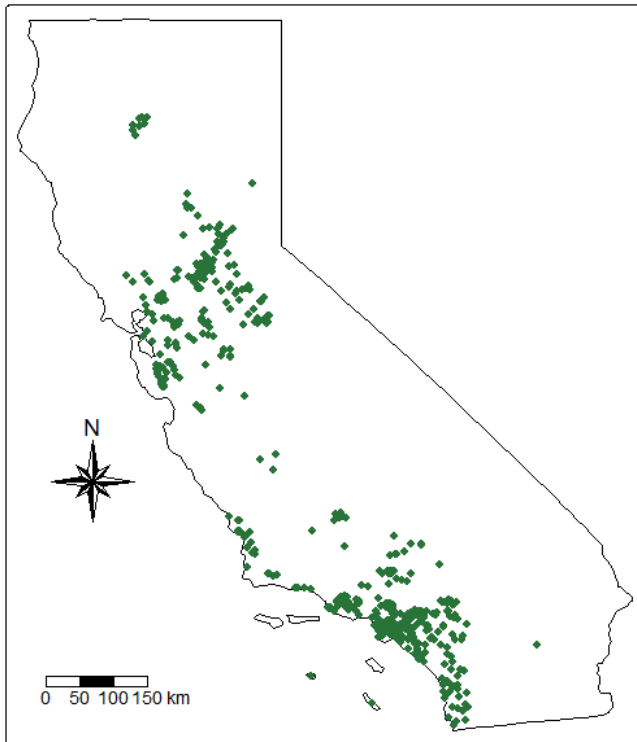


Figure 2. Community water systems ($n=475$) associated with the 74 locations of interest analyzed for the community surface water reliability key outcome.

Community surface water reliability (key outcome)

Description

To capture a metric of community surface water delivery reliability considering both the frequency and severity of instances when deliveries do not meet drinking water demands, we compared surface water deliveries modeled in CalSim3 (“M&I surface water deliveries”) to estimates of recent potable water demands associated with the CalSim3 modeled supplies (“estimated recent potable demands”). We then categorized the 74 LOIs into outcome levels according to two key water supply reliability criteria: 1) How frequently deliveries cover current estimated potable demands (reliability), and 2) the extent to which there are years with large or critical shortfalls in surface water deliveries (severity).

Importantly, in identifying M&I water users and calculating current estimated potable demands, we only include those utilizing CalSim3 modeled supplies for direct potable uses. Thus we exclude users who exclusively utilize these surface water sources for conjunctive uses (groundwater recharge and/or Aquifer Storage and Recovery (ASR)) and all demands associated with conjunctive uses (see LOIs section above). As such, the community surface water delivery reliability outcome levels only consider the sufficiency of surface water deliveries

to meet estimated current direct potable demands for those specific sources. Additionally, the outcome levels do not address other water supplies utilized in parallel with these sources and thus do not fully assess the potential for supply shortages among M&I users. Depending on the other water sources available to a M&I water user and/or whether they have long-term storage to store excess deliveries in wet years, the impact of the reliability measure will have a different effect on the overall water supply of a given user. For example, through additional or alternative supplies and/or through water storage, some providers can compensate for even extreme variation in surface water deliveries avoiding any shortfalls in their overall supply portfolio. In this case being in Level 3 or Level 4 may not be an issue. Conversely, a provider that is fully reliant on these supplies without long-term storage may see impacts to their overall supply with small changes in reliability. In this case, dropping from Level 1 to Level 2 could have a very detrimental effect.

Key Outcome Levels

Discrete outcome level calculation

The community surface water delivery reliability outcome levels (1-4, integer) provide a high-level snapshot of M&I surface water delivery reliability considering both the frequency with which estimated recent potable demands are satisfied across the 100-year period of each CalSim3 scenario (i.e. reliability), and when not satisfied, how big the difference is between modeled deliveries and estimated recent potable demands (i.e. severity). The four outcome levels and their unique definitions are described in Table 1.

Table 1. Outcome levels descriptions and definitions

Outcome level	Description	Reliability Criteria	Severity Criteria	Continuous spread within outcome level
1 (Optimal)	Surface water deliveries cover estimated recent potable demands for surface water (excluding demands associated with recharge or Aquifer Storage and Recovery (ASR)) in most years, no years with large shortfalls	90% or more of estimated recent potable demands met in at least 90% of years	For years with lower than 90% of estimated recent potable demands met, deliveries are at least 70% of estimated recent potable demands	‘Drop risk’ calculated by selecting the worst of: 1) reliability score, and 2) severity score, both rescaled between 0-0.99 based on the lower & upper thresholds of the respective outcome level and criterion.
2 (Acceptable)	Surface water deliveries cover estimated recent potable demands for surface water (excluding demands associated with recharge or Aquifer Storage and Recovery	90% or more of estimated recent potable demands met in at least 50% of years	For years with lower than 90% of estimated recent potable demands met, deliveries are at least 50% of estimated recent	

	(ASR)) in at least half of years, no years with critical shortfalls		potable demands	
3 (At Risk)	Surface water deliveries cover estimated recent potable demands for surface water (excluding demands associated with recharge or Aquifer Storage and Recovery (ASR)) in at least half of years, critical shortfalls occur in no more than 20% of year	90% or more of estimated recent potable demands met in at least 50% of years	For years with lower than 90% of estimated recent potable demands met, no more than 20% of years with deliveries less than 50% of estimated recent potable demands	
4 (Critical)	None of the above criteria met	None of the above criteria met	None of the above criteria met	

Continuous outcome level calculation

We additionally expanded these discrete outcome levels into a continuous outcome level framework, where we kept the above outcome level assignments, but added a continuous spread within each outcome level as follows.

- 1.00–1.99 for Level 1
- 2.00–2.99 for Level 2
- 3.00–3.99 for Level 3
- 4.00–4.99 for Level 4

For example, LOIs in Level 1 could have a value between 1.0 and 1.99, yielding an overall range of outcome level values 1.0 to 4.99. Within each existing outcome level, we assigned continuous values as follows: first, we calculated rescaled (between 0-1) scores for the reliability and severity metrics. For the reliability metric, this meant rescaling the percentage of years in which demand was met based on the upper and lower thresholds for that respective outcome level. See the equation below as an example for Level 2.

$$\text{Level 2 reliability score} = \frac{\% \text{ years where } \geq 90\% \text{ of demand is met} - \text{level 2 reliability threshold (ie: 50\%)}}{\text{level 1 reliability threshold (ie: 90\%)} - \text{level 2 reliability threshold (50\%)}}$$

In the generic formula of

$$\text{Level X reliability score} = \frac{A - B}{C - D}$$

the formula takes on the following terms for *A*, *B*, *C*, and *D* at each discrete outcome level. *A* is always the percentage of years in which at least 90% of demand is met. *B* and *D* are 90%, 50%, 50%, and 0% for Level 1, 2, 3, and 4, respectively, to align with the lowest reliability threshold possible at that outcome level. *C* is 100%, 90%, 90%, and 50% for Level 1, 2, 3, and 4, respectively, to align to the threshold necessary to improve to the next outcome level or the maximum possible (100%) in the case of Level 1.

Similarly, finding the severity score required rescaling the worst year's percentage of demand met based on that outcome level's upper and lower thresholds. See the equation below for an example for Level 2.

$$\text{Level 2 severity score} = \frac{\% \text{ demand met in worst year} - \text{level 2 severity threshold (ie: 50\%)}}{\text{level 1 severity threshold (ie: 70\%)} - \text{level 2 severity threshold (50\%)}}$$

For Level 1 severity score calculations, we use 100% as the upper threshold and 70% for the lower thresholds to align with Level 1 upper and lower thresholds.

For Level 3 and 4 severity score calculations, we use structurally different formulas that use the percentage of years in which demand met is below 50% to align with how those outcome levels are defined. The Level 3 and Level 4 formulas for severity score are below.

$$\text{Level 3 severity score} = \frac{\text{Level 3 severity threshold (ie 20\%)} - \% \text{ of years with demand met below 50\%}}{\text{Level 3 severity threshold (ie 20\%)}}$$

$$\text{Level 4 severity score} = \frac{100\% - \% \text{ of years with demand below 50\%}}{100\% - \text{Level 3 severity threshold (ie 20\%)}}$$

For each LOI, we then converted these two normalized scores into 'drop risk' scores: ie: a given location's proximity of dropping down a level based on poor performance based on its reliability or severity metrics:

$$\text{Reliability drop risk} = (1 - \text{normalized reliability score}) * 0.99$$

$$\text{Severity drop risk} = (1 - \text{normalized severity score}) * 0.99$$

We then took the lowest of these two normalized scores as the 'drop risk'. For example, a location in Level 3 could have demand met in 78% of years (leading to a rescaled reliability score of 0.70 and a reliability drop score of 0.297) but be below 50% of demand in 19% of years (leading to a rescaled severity score of 0.05 and severity drop score of 0.94). This high severity drop score indicates that this location would be quite close to dropping down into Level 4 (since Level 3 requires no more than 20% of years with deliveries <50% of demand).

Finally, we integrated this 'drop risk' score into a continuous outcome level value as follows:

$$\text{Continuous outcome level value} = \text{discrete outcome level} + \text{highest drop risk}$$

ensuring that all locations stay within their original outcome level, and that high decimals within that outcome level (ie: high drop risk) indicate proximity to dropping into a worse outcome level, while low decimals within a given outcome level (ie: low drop risk) indicate proximity to improving an outcome level.

Figures 3 and 4 display baseline scenario results for both the categorical and continuous versions as described above. Results from the baseline scenario for the two distinct calculations used to develop the continuous outcome level values are displayed in Figure 5 and Figure 6. As shown, the continuous outcome level value for the most part increases alongside the percentage of years in which deliveries meet recent potable demands. However, there are some exceptions: for instance, some LOIs in Level 3 may have surface water deliveries meet their recent potable demands more frequently than some of the locations in Level 2. In these cases, the locations in Level 3 would have been assigned a lower outcome level because in at least one of the years in which demand wasn't met, that location suffered a critical shortfall (ie: deliveries were less than 50% of recent potable demands).

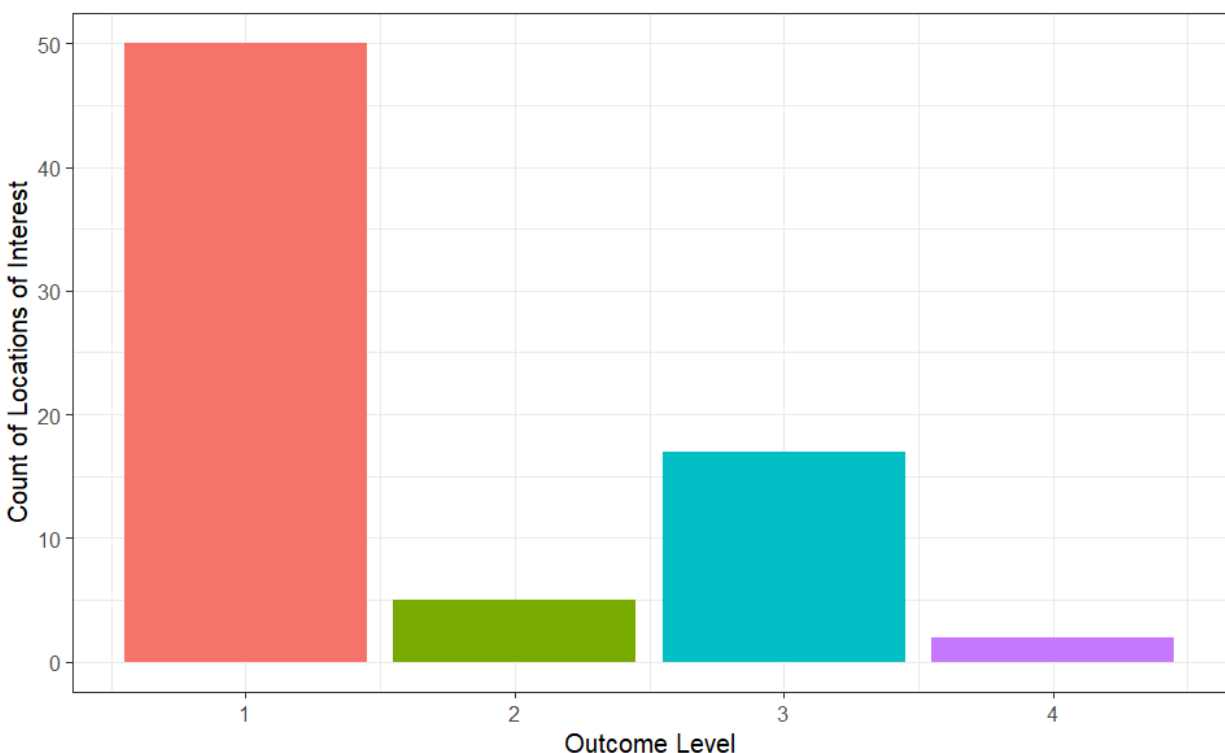


Figure 3. Distribution of categorical outcome level outcomes for all relevant locations using the baseline scenario (s0020).

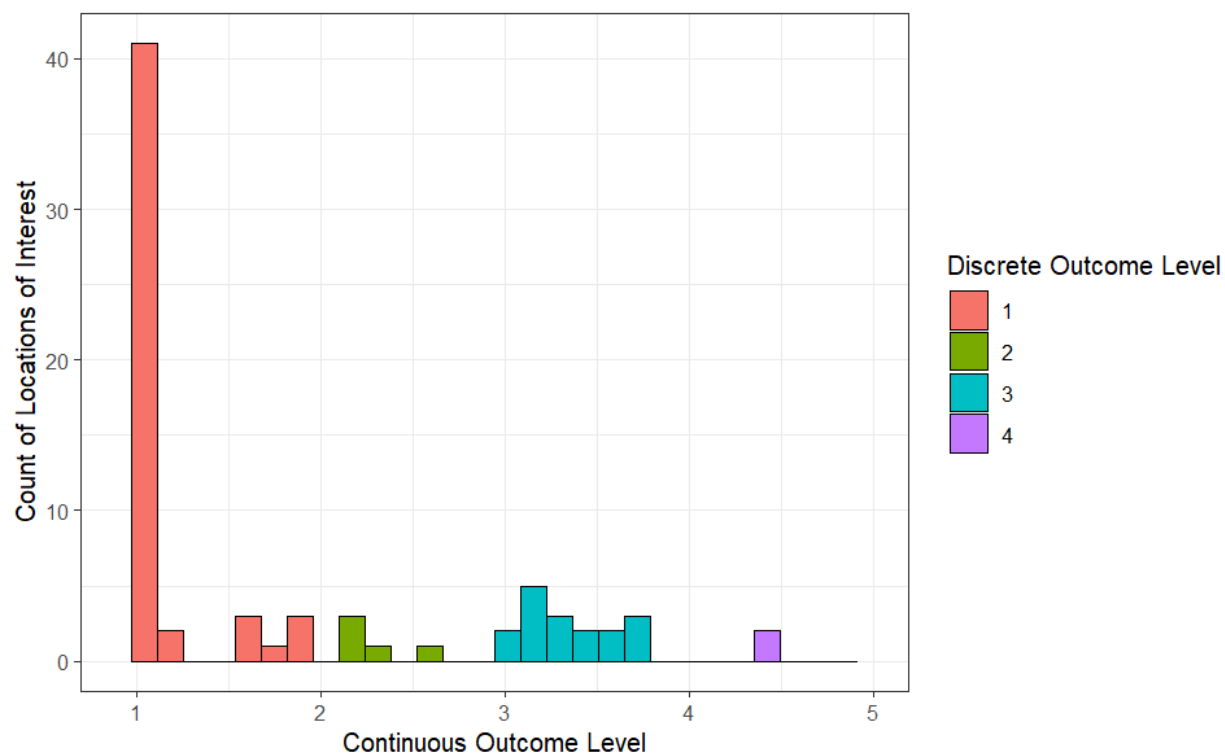


Figure 4. Distribution of continuous outcome level outcomes for all relevant locations using the baseline scenario (s0020).

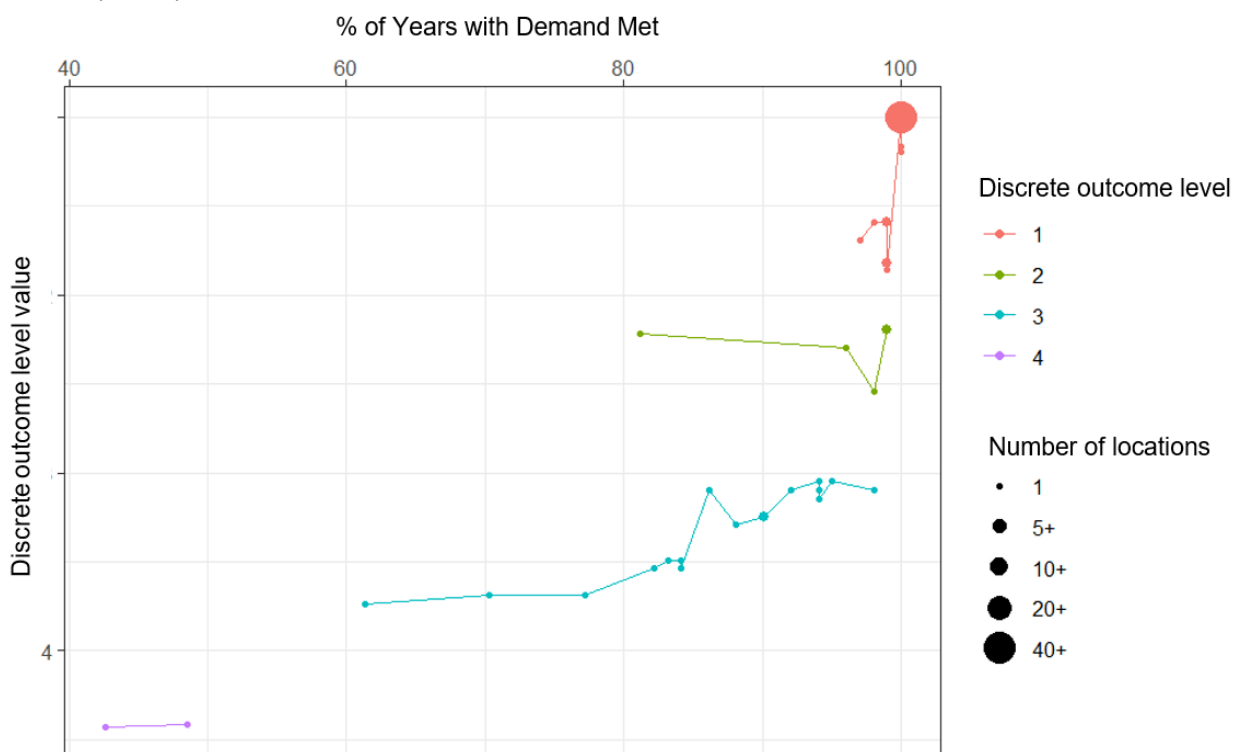


Figure 5. Distribution of the percentage of years in which demand is met (ie: deliveries equal 90% or greater of recent potable demands) across discrete and continuous outcome levels in the baseline scenario (s0020).

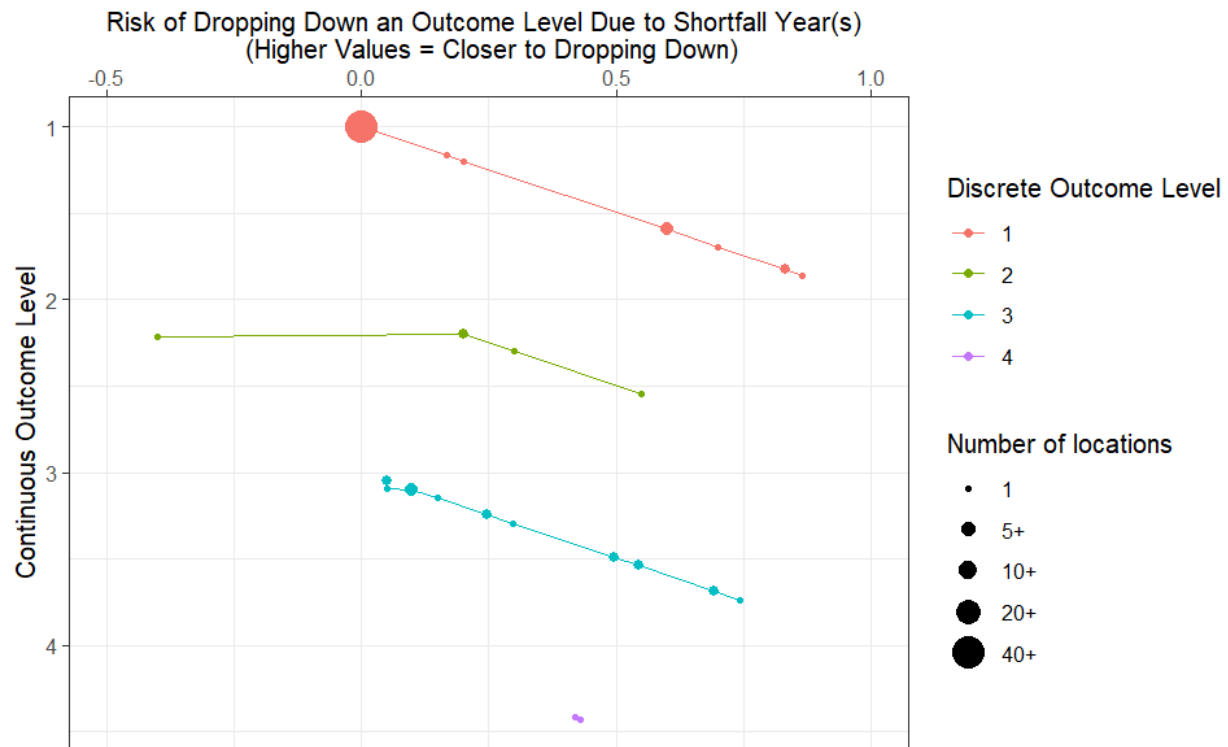


Figure 6. Severity drop risk across discrete and continuous outcome level values for locations of interest in the baseline scenario (s0020). Severity drop risk = $(1 - \text{normalized severity score}) \times 0.99$.

Methodology

The community water delivery outcome levels were calculated in three distinct steps. First, we estimated recent potable demands for each LOI using data reported from community water systems to the State Water Resources Control Board's Division of Drinking Water. We then calculated annual surface water deliveries for each LOI using CalSim3 data extractions for each scenario. Finally, we compared the modeled surface water deliveries to empirical demands in order to evaluate to what extent deliveries meet drinking water demands.

Step 1: Estimates of recent potable demands for each location of interest.

In order to estimate potable demands for the surface water sources modeled in CalSim3, we needed to identify the water systems that utilize these sources. We did this using the surface water sources, sales and transfers for California community water systems dataset (Dobbin et al., 2026) which identifies the surface water sources of 849 community water systems, as well as 3 non-transient non-community water systems and 6 urban water suppliers that are not

public water systems but provide water to one or more community water systems. Surface water sources include streams, springs, lakes, canals, rivers, PWS, non-PWS wholesaler, and other suppliers (defined as entities that are not public water systems or wholesale drinking water providers but that sell water to community water systems, such as local governments and independent special districts).

Using that dataset, we reviewed each source that was not purchased from another water system (n=364) and compared that information to the CalSim3 report (Department of Water Resources, 2022) to determine if each of these unique water demands were represented within CalSim3 or not. Examples of non-represented sources that were designated as “out of CalSim3” include springs, local creeks, independently operated reservoirs, etc. Conversely, examples of represented sources include all SWP and CVP contracts. For all represented sources, we assigned a LOI or diversion node as noted in the CalSim3 report (e.g. 12_NU1; ACFC; SBA029). In a few select instances, the report did not include the specific diversion node and was rather listed as “multiple nodes” (e.g. Metropolitan Water District) (Department of Water Resources, 2022 p. 14-28). In these cases, a placeholder diversion node code was created to represent these water demands. Next, to account for purchased sources, we carried forward the demand unit assignments to all purchased water sources, thereby allocating the percentage of each system’s total water supply across all LOIs that potentially serve it. This resulted in each row in the dataset being assigned to a specific LOI or diversion node, to groundwater, or designated as “out of CalSim3”. For instance, CA0110003 has 65.0% of its volume come from the ACFC LOI, 11.4% from out of CalSim3, and 23.6% from groundwater. Lastly, we reviewed these demand unit assignments against the CalSim3 report (Department of Water Resources, 2022) and reassigned purchased sources if they were represented differently in CalSim3 to align as much as possible with the existing model. For example, California State Prison – Solano has its own demand unit in CalSim3 (CSPSO) which we utilized instead of the original assignment of two demand units, Solano Irrigation District and the City of Vacaville, from whom the water is purchased.

We then multiplied these percentages by total potable demands for each system derived from the SAFER clearinghouse data, resulting in an estimate of recent potable demands by source for each system in acre-feet. Data on recent potable water demand comes from the SAFER clearinghouse data (2025) and is defined as “total volume of potable water used by both residential and non-residential users during a calendar year” with a reporting period spanning 2022 to 2025, depending on the system. For systems missing this data (310 of 474 systems), we imputed the median demand for the hydrologic region a given system falls within (obtaining [hydrological region shapefiles from](#) the Department of Water Resources) (*i03 Hydrologic Regions*, n.d.). These system-level demands (in TAF) were then summed for each of the 74 LOIs representing estimated recent potable demands by demand unit or diversion node.

Step 2: Calculation of surface water deliveries by location of interest and scenario.

The next major step was to calculate annual surface water deliveries (in TAF) by LOI for all of the scenarios. Working with the COEQWAL modeling team, we developed a [master crosswalk](#) identifying delivery arcs representing M&I water deliveries within CalSim3 for each of the 74 LOIs. These delivery arcs include only M&I surface water deliveries. Project carryover deliveries were included where applicable. Using CalSim3 data extractions for surface water deliveries provided by the modeling team, we calculated monthly deliveries (in TAF) by LOI and modeling scenario. Finally, we aggregated deliveries by year and calculated the percentage of estimated recent potable demands that each year's deliveries represented. These annual calculations are available in the data-in-depth explorer for use, as are monthly deliveries calculated using the same methods. Also available in the data-in-depth tool is a scenario average for deliveries by LOI ('long-term delivery average'). Notably, for consistency purposes the average excludes partial calendar years (1921 and 2021) so spans 1922-2020.

Step 3: Comparison of estimated recent potable demands and surface water deliveries by location of interest and assignment of categorical and continuous outcome level values.

For each LOI and scenario, we calculated the percentage of years in which deliveries fully or nearly-fully met (which we defined as 90% or more) recent potable demands. To capture the frequency and severity of years with large or critical shortfalls, we also calculated the percentage of years in which deliveries were at least 70% and 50%, respectively, of recent potable demands. Finally, for each modeling scenario, LOIs were categorised into outcome levels based on these metrics. See Table 1 above for the formal outcome level definitions.

Guidelines for interpretation

It is important to be clear about what is and is not captured in the community surface water delivery reliability metric. This metric considers only the reliability of surface water deliveries modeled in CalSim3 (primarily State Water Project and Central Valley Project supplies). Other sources of supply including groundwater, most local surface water supplies and desalinated water are not included. Secondly, the metric only captures direct potable demands excluding water deliveries utilized for groundwater recharge or aquifer storage and recovery. Both are important sources of supply for community water systems across the state that are unaddressed herein. **Thus the reliability metric is source specific and does not address the reliability of a community water system or wholesale provider's cumulative water supply or provide any indication of their ability to provide reliable water supplies to its customers.** Whether and to what extent varying reliability of these specific water supplies will have an impact on the overall water supply of a drinking water provider depends on the provider. Through additional or alternative supplies and/or through water storage, some providers can compensate for even

extreme variation in surface water deliveries avoiding any shortfalls in their overall supply portfolio. Conversely, a provider that is fully reliant on these supplies without long-term storage may see impacts to their overall supply with small changes in reliability.

Considering this focus the community water delivery outcome levels are best suited to identifying conditions under which modeled surface water supplies are more or less reliable by comparing between scenarios. This can be done on aggregate using all LOIs, or at specific LOIs representing specific groups of M&I users. Due to the nature of its calculation, the reliability metric attempts to provide a single data point summarizing the variation in the deliveries across the entire scenario period (100 years). Other metrics available in the data in depth tool, provide more detailed results for specific LOIs that can be broken down by year or month and may be more suitable for analyzing individual LOIs with more nuance.

Limitations

There are several limitations with the community surface water deliveries reliability metric that should be noted. First, to contextualize the drinking water implications of variable M&I surface water deliveries, the reliability metric compares deliveries against estimated recent potable demands for that specific source based on recent usage. These estimates are imperfect and rely on imputation where data gaps on recent usage exist which affects a large number of systems (n=310) and in turn a large number of LOIs (n=63). Relatedly, recent usage of a source is not always a good predictor of future use or need. The water supply sources of a community system and the degree of their reliance on any given source can change year to year. For example, the loss of a well can lead to increased reliance on surface water, or vice versa, installation of new wellhead treatment or the construction of new wells can reduce surface water reliance. Lastly, CalSim3 assigns static water demands across each 100-year scenario based on recent demands. Whether demands will change in the future is an important consideration for understanding the implications of the results.

Code and data sources

- Code packages and availability
 - All analysis was conducted in RStudio (R version 4.4.1, RStudio 2026.04.0) specifically using the tidyverse, readxl, and sf packages
 - All source code is documented in this [github](#) repository
- Data sources
 - Outputs from CalSim3 model runs
 - Dobbin, K., Zamora-Olivares, A., Cohen, M., Reade-Malagueño, B., Rempel, J., & Gomez-Cervantes, A. (2026). Surface water sources, sales, and transfers for California community water systems [Dataset]. Dryad. <https://doi.org/10.5061/dryad.z08kprw2>
 - *CA SDWIS (Safe Drinking Water Information System)*: State Water Resources Control Board Drinking Water Watch dashboard (<https://sdwis.waterboards.ca.gov/PDWW/>). Query for all active public water systems. Retrieved November 18, 2025.

- *DDW System Area Boundary Layer (SABL)*. (n.d.). [Dataset]. California Water Boards. Retrieved May 11, 2026, from <https://gispublic.waterboards.ca.gov/portal/apps/experiencebuilder/experience/?id=7f249a13de9f48e084167e1ed783278b>
- *I03 Hydrologic Regions*. (n.d.). [Dataset]. California Open Data Portal. Retrieved May 11, 2026, from <https://data.ca.gov/dataset/i03-hydrologic-regions>
- *SAFER clearinghouse* (2025). 2025 SAFER clearinghouse dataset received directly from the State Water Resources Control Board Division of Drinking Water on April 24, 2025.

References

Department of Water Resources. (2022). CalSim3 Report: A water resources system planning model for state water project & central valley project. Available at: https://s3.amazonaws.com/og-production-open-data-cnra-892364687672/resources/2d4160d7-cbe1-4e63-8cdd-98f322e74cf2/calsim-3-report-final.pdf?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAJJIENTAPKHZMIPXQ%2F20260728%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20260728T220531Z&X-Amz-Expires=86400&X-Amz-SignedHeaders=host&X-Amz-Signature=b077f16b1f4e4dd43bfed955d5daff778c7144df781703434fd0c06ba738f575

Additional metrics

As discussed above, the community surface water reliability key outcome only considers project deliveries modeled in CalSim3 and therefore cannot address the potential for water shortages among drinking water providers or the potential economic implications of these shortages. To address these gaps we developed two additional metrics available in the data-in-depth explorer that consider M&I surface water deliveries alongside M&I groundwater pumping at 83 LOIs within the geographic domain of the CalSim3 model (the Central Valley). These two metrics, M&I supply shortage and welfare loss from supply shortage, and their associated methods are described below.

M&I supply shortage

Description

In addition to the community delivery outcome that focuses on surface supplies, we define an additional outcome – the M&I supply shortage – that estimates the reduction of supply deliveries with respect to total supplies (including both surface and groundwater deliveries). By including both surface and groundwater supply deliveries we can consider the importance of the surface supply reduction in the water user supply portfolio: for instance, while for a water user that relies

mostly on surface supplies, a surface supply reduction can be really important, for a water user mostly reliant on groundwater supplies, a reduction of surface deliveries can be insignificant.

Locations of interest

Groundwater demands in CalSim3 are only modeled within the Central Valley corresponding with CalSim3's spatial extent. As such the LOIs for these outcomes consist exclusively of CalSim3 demand units within the model's geographic domain with M&I demands (whether surface water, groundwater or both). There are 83 LOIs in this subset as displayed in the ["Master list of M&I supply locations of interest"](#) datasheet. This sheet also provides the latitude and longitude associated with the centroid of these CalSim3 polygons for spatial representation as displayed in Figure 3.

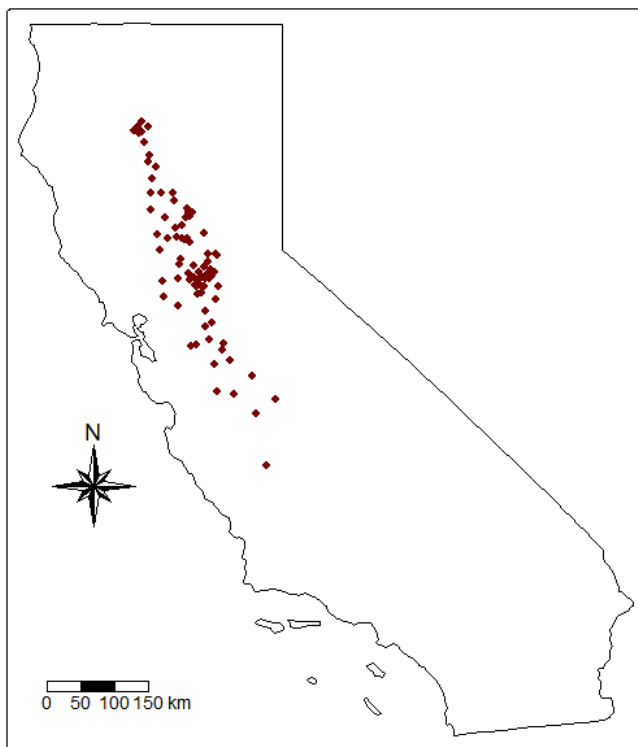


Figure 7. Spatial distribution of the M&I supply locations of interest (n=82 with one demand unit missing a spatial reference point).

All results are presented at the LOI level and therefore do not directly correspond to results for any specific water system. However, to aid users in identifying results of particular relevance for specific communities, we also identify as many of the California-regulated community water systems associated with each of these LOIs as possible. We did this using the CalSim3 report and triangulating the listed water providers with the State Safe Drinking Water Information System (SDWIS) to associate each system with a unique public water system IDs where possible. This yields more than 160 communities associated with 150 unique water systems as displayed in the ["Master list of systems served by M&I supply locations of interest"](#) datasheet. These systems are mapped in Figure 8 using their centroids from the System Area Boundary

Layer dataset (with spatial data missing for four systems). We provide these water systems as helpful context for understanding the results from outcomes four and five however we stress that all results are at the demand unit level and cannot be applied directly to any of the associated water systems individually.

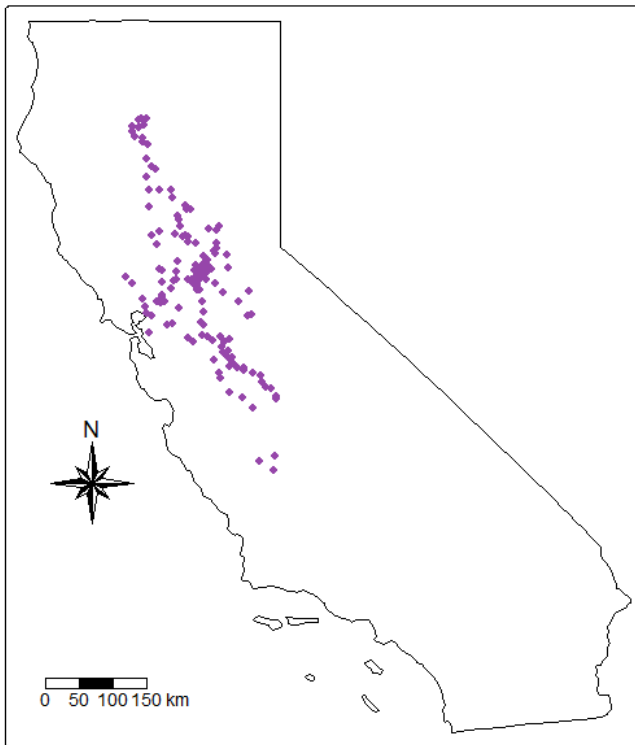


Figure 8. Map of water systems associated with M&I supply locations of interest with spatial data available ($n=213$).

Methodology

Estimating M&I supply shortages involve two steps. First, we obtain surface water and groundwater deliveries to the LOIs from CalSim3. Second, we estimate water shortages – that is, periods when water deliveries are insufficient to meet demand – for each LOI.¹

Step 1: Estimating Water Deliveries.

CalSim3 provides monthly surface and groundwater deliveries for each LOI. In the following figure (Figure 9), we show the overall surface vs groundwater supply reliance for each LOI in the map on the left calculated as the average annual deliveries (from one or other source) with respect to the total deliveries. As we can see, some LOIs are mostly reliant on surface water, others have a mix of surface and groundwater supplies in their portfolio, while some other LOIs are mostly or exclusively reliant on groundwater.

¹ This methodology is analogous to the estimation of water supply shortages for the agricultural LOIs

Importantly, deliveries in CalSim3 change annually. First, demands are influenced by temperature (or evapotranspiration) and precipitation. Wetter years tend to have lower demands (as some of the outdoor water use comes directly from precipitation), while hotter years result in increased demands and need greater applied water. Second, in CalSim3, surface supplies vary to represent water availability and operation rules in the Central Valley. That means that during drier years, surface water supplies will be reduced – this reduction will be more acute for those users with junior water rights, or less seniority. A key assumption in CalSim3 is that demands are met even in these drier years by increasing groundwater pumping. Therefore, as shown in the three LOI examples on the right of Figure 9, when surface deliveries decline because of droughts, groundwater deliveries will increase, mitigating all potential shortages.

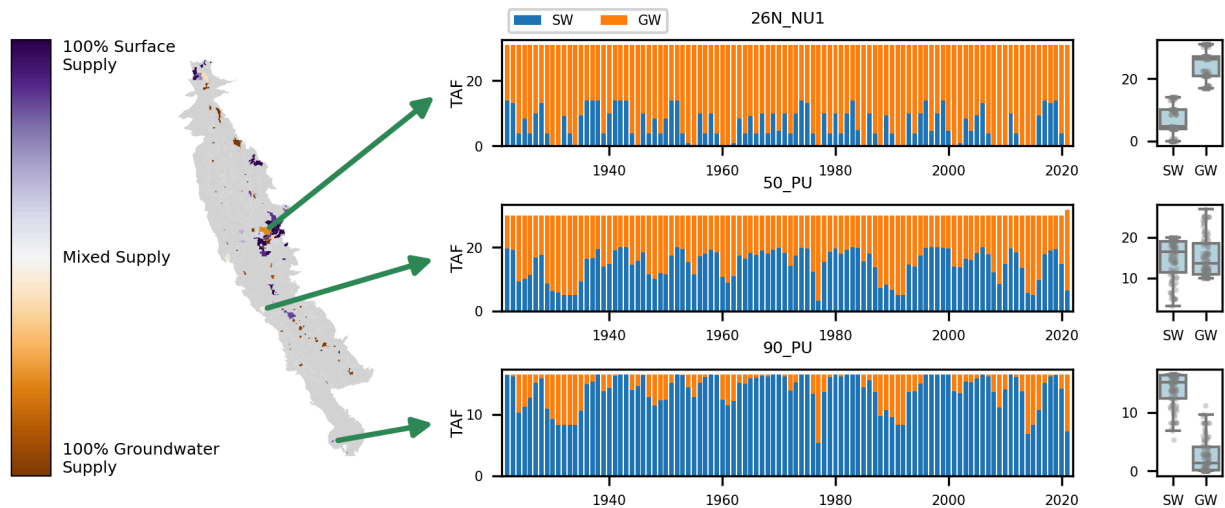


Figure 9. Spatial distribution of water supply reliance across demand units (left). Darker purple shades indicate greater reliance on surface water, while darker brown shades indicate stronger dependence on groundwater. Selected examples of LOIs (right) illustrate historical variability in surface water (SW) and groundwater (GW) supplies for representative demand units, highlighting differences in supply composition and variability over time.

While groundwater pumping generally increases during droughts, M&I users are not always able to fully compensate for reduced surface water deliveries through groundwater pumping alone. During severe drought periods, groundwater limitations and economic constraints can force users to reduce water use. At this stage, shortages experienced by individual users are not directly represented within the CalSim3 model, representing an important limitation.

Step 2: Estimating M&I Water Shortages

To overcome this limitation, we estimate water shortages as reductions in surface water deliveries relative to baseline conditions, assuming that groundwater pumping increases to compensate for the reduction in surface deliveries, relative to total water demands. When surface water deliveries fall below what would be considered normal (i.e., non-drought) conditions and groundwater pumping exceeds typical levels, the increase in groundwater pumping is interpreted as evidence that users are compensating for reduced surface water supplies, thereby indicating a surface water shortage. The second condition, the increase in groundwater pumping, is an important condition because in years with lower demands (for instance during wet years) surface deliveries might decrease without any pumping increase. Following our assumption, this does not represent a shortage, just a reduction in demands.

Mathematically, to represent the full level of available surface supply for non-drought years, a benchmark for surface-water deliveries is defined as the 75th percentile of surface deliveries – or “net diversion” in CalSim3 language. We prefer to use the 75th percentile instead of the maximum value because the latter may reflect deliveries during exceptionally wet years, which would overstate the typical level of available surface supply. In addition, groundwater pumping must exceed its median level (50th percentile) for a year with reduced surface deliveries to be classified as a shortage. The median pumping level represents normal groundwater use, as many users rely on groundwater for a baseline share of their supply even during non-drought years. An example of this classification is shown in Figure 10.

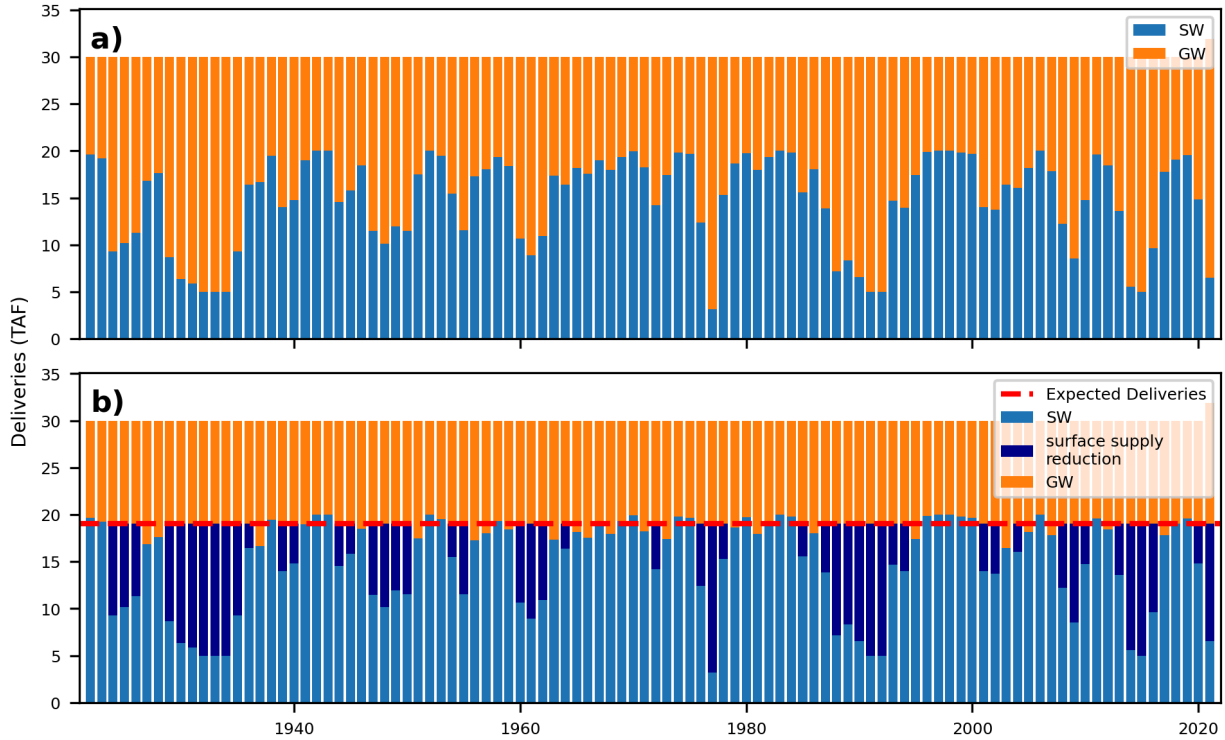


Figure 10. Example of a demand unit with a mixed water supply from surface water and groundwater. Surface water deliveries are shown in light blue, groundwater pumping in orange, and the dark-blue portion represents the reduction in surface deliveries relative to the benchmark (interpreted as shortage). The red line represents the 75th percentile benchmark of surface-water deliveries used to define non-shortage surface supply levels.

The final shortage will be obtained as the ratio of the reduction in surface supplies with respect to the total water demands – obtained as initial total supplies from surface and groundwater.

$$Shortage (\%) = \frac{\Delta(SW)}{D} = \frac{\Delta(SW)}{SW + GW}$$

Where:

- ΔSW = surface supply reduction
- SW = surface water supply
- GW = groundwater supply
- D = total demand

To compare scenarios consistently, expected deliveries are first calculated from the baseline scenario (Scenario 0020) as the sum of surface deliveries and shortages for each year. This

represents the level of surface deliveries that would have occurred in the absence of shortages and serves as a reference baseline for comparison across scenarios.

For the alternative operational and climate change scenarios, shortages are identified when surface deliveries fall below the expected delivery baseline and the deficit is compensated through additional groundwater pumping (exceeding 50th percentile of the baseline scenario). This approach allows shortages to be quantified relative to baseline operating conditions while accounting for increased groundwater reliance under more severe drought conditions.

See the Agricultural Revenue: Description and Documentation for a more detailed description of the analysis of shortages under different climate and regulatory scenarios.

Guidelines for interpretation

This metric should be interpreted as the impact of the reduction of surface deliveries with respect to the total deliveries (from surface and groundwater). Therefore, the M&I shortage metric does not represent actual unmet demands experienced by users, but rather reductions in surface water deliveries relative to total water supply. As a result, shortages are strongly influenced by the composition of users' water supply portfolios, particularly their dependence on surface water versus groundwater supplies.

Users that rely almost entirely on groundwater supplies may experience little to no modeled shortages because reductions in surface deliveries have limited influence on their total supply portfolio. This was observed during the pre-SGMA period, where groundwater-dependent users experienced limited drought impacts within the framework. In contrast, users that rely heavily on surface water deliveries may experience larger shortages when surface supplies decline substantially during dry periods. However, some surface-water-dependent LOIs possess more senior water rights and therefore experience relatively small reductions in deliveries, resulting in limited shortages despite their dependence on surface water supplies. Similarly, LOIs that primarily depend on groundwater but also receive supplemental surface deliveries may experience only minor shortages because groundwater remains their dominant source of supply.

Another caveat is how users are represented through LOIs. Results are evaluated at the LOI (demand unit) scale used in CalSim3, which aggregates multiple users and regions into larger operational units. Therefore local-scale differences in groundwater access or infrastructure may not be fully captured.

Limitations

There are some important assumptions in the estimation of this metric. First, we are assuming that any surface supply reduction with groundwater compensation becomes a water shortage. That implies that all groundwater pumping increases are considered shortages. This assumption can be considered a "worst case" scenario because some of this groundwater compensation might not be unsustainable over the long-term (this is especially important in groundwater basins that don't present long-term overdraft, but even in overdrafted basins some groundwater pumping might be sustainable). Another important assumption is the definition of the thresholds: we chose the 75th percentile for the "normal or non-drought" surface supply deliveries and the

50th percentile for the “normal or non-drought” groundwater pumping. While we checked these metrics for consistency across all LOIs, there might be differences in local conditions that would suggest using different metrics.

Code data sources:

- Code and availability
 - All analysis was done on Python.
- Data sources:
 - Outputs from CalSim3 model runs

Welfare loss from supply shortage

Description

To measure the economic impacts of the above described M&I water supply shortage outcome, we estimate the welfare loss for residential consumers at each of the relevant 83 LOIs for all of the CalSim3 scenarios using the modeled supply shortage and a calibrated constant-elasticity demand function. Welfare loss measures the value consumers would have been willing to pay above prevailing water rates to avoid the water reduction they face in any given scenario. As such, welfare loss provides a policy-relevant monetary measure of shortage burden suitable for comparing scenarios, evaluating drought management alternatives, and identifying communities most exposed to supply restrictions. The metric captures consumer surplus losses to residential water users and is intended to be interpreted alongside the M&I supply shortage results, which provide the modeled shortage volumes that drive the welfare loss calculation.

Outcomes

The welfare loss outcome reports annual residential consumer welfare loss in dollars per year for each of the M&I supply LOIs. Figure 11 shows an example of annual welfare losses under baseline conditions (s0020) across four reaggregated LOIs: Sacramento River, San Joaquin River, Sacramento River/San Francisco Bay, and Tulare Lake. The shaded gray bands highlight selected drought periods. The results show that welfare losses are highly uneven across years and regions. Most years have relatively low losses, but several dry years generate very large spikes.

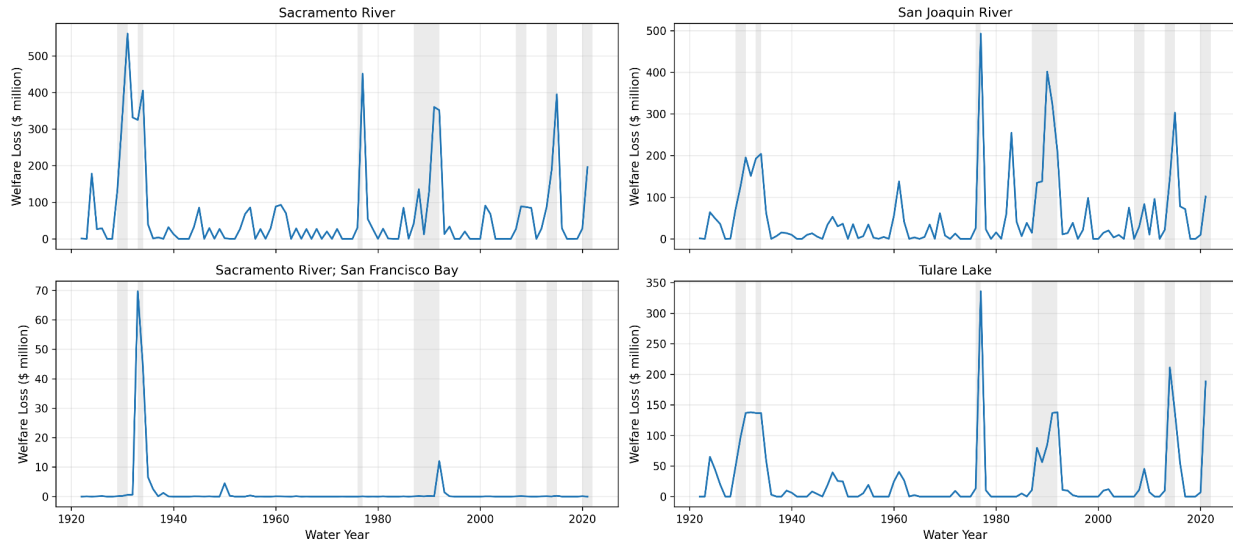


Figure 11. An example of annual welfare loss under the baseline scenario (s0020) by aggregated locations of interest.

Locations of interest

The LOIs are the same as the M&I supply shortage outcome described above: 83 CalSim3 demand units with M&I surface water deliveries and/or M&I groundwater pumping.

Methodology

Step 1: Compile water price and elasticity data and calibrate the inverse demand function

For each LOI, we compiled publicly available information on the prevailing volumetric tier price and fixed monthly charge from utility rate schedules at the community water system level wherever available. Data was available for 94 systems across 63 demand units. All prices are standardized to 2020 dollars and expressed in dollars per hundred cubic feet (\$/ccf) at the 12-ccf monthly residential billing benchmark (Ayres et al., 2021) and then weighted by service population to obtain a representative price for the LOI as a whole. We assume a constant price elasticity of residential water demand of $\varepsilon = -0.4$, drawn from California-based estimates summarized in Bruno & Jessoe (2021). Let Q^{Full} denote the baseline full demand for the LOI and P the prevailing marginal tier price, we calibrate a constant-elasticity inverse demand function:

$$P(Q) = P \times \left(\frac{Q}{Q^{Full}} \right)^{\frac{1}{\varepsilon}}$$

Step 2: Compute annual welfare loss under each CalSim3 scenario.

For each LOI (with available data), scenario, and year, we utilize the corresponding supply shortage from M&I supply shortage metric described above. We then define the realized residential consumption Q^R as the delivery net, and integrated the inverse demand function from Q^R up to Q^{Full} to obtain the annual welfare loss:

$$WL(Q^R) = \left(\frac{\varepsilon}{1+\varepsilon}\right) \times P \times Q^{Full} \times \left[1 - \left(\frac{Q^R}{Q^{Full}}\right)^{\left(\frac{1+\varepsilon}{\varepsilon}\right)}\right]$$

When Q^R equals or exceeds Q^{Full} (no shortage) welfare loss is set to zero. The functional form is convex, each additional unit of water cutoff commands a sharply rising marginal willingness to pay.

Guidelines for interpretation:

The reported outcomes should be interpreted as dollar-value losses caused by water supply shortages under specific scenarios. These losses reflect the value of foregone water use to residents, measured by their willingness to pay to avoid reductions in water supply.

Limitations:

In addition to the limitations described for the M&I shortage metric, which also affect the welfare loss calculation, there are two additional limitations to note. First, this analysis estimates welfare loss from the consumer side only and does not include producer surplus losses. As a result, the welfare loss reported in this outcome should be interpreted as consumer welfare loss, reflecting consumers' willingness to pay to avoid reductions in water supply.

A second limitation is that the analysis focuses on water quantity rather than water quality. The willingness-to-pay framework compares desired water demand with actual available demand under shortage conditions, treating the gap as a quantity-based shortage. It does not capture additional willingness to pay associated with water quality concerns, such as households purchasing bottled water to obtain safer or higher-quality drinking water. Therefore, this method captures welfare losses related to reduced water deliveries only.

Code and data sources:

- Data sources
 - M&I supply shortage metric as described above
 - Residential tiered water price data, collected from online documentation by water utility and assigned to the relevant LOI as described in methodology for the key outcome above. Available upon request.

References

Ayres, A., Hanak, E., McCann, H., Mitchell, D., Sugg, Z., & Rugland, E. (2021). *Groundwater and Urban Growth in the San Joaquin Valley*. Public Policy Institute of California.

Bruno, E. M., & Jessoe, K. (2021). Using Price Elasticities of Water Demand to Inform Policy. *Annual Review of Resource Economics*, 13, 427–441.